

# Title of the Script

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# Chapter 1

## Limits

### Definition 1.1 Limit

Let  $f : D \rightarrow \mathbb{R}$  be a function with  $D \subseteq \mathbb{R}$  and  $a$  a cluster point of  $D$ . Then  $L$  is the limit of  $f$  at the point  $a$ , if for all  $\varepsilon > 0$  there exists a  $\delta > 0$ , such that for all  $x \in D$  with  $0 < |x - a| < \delta$  we have:  $|f(x) - L| < \varepsilon$ . We write:

$$\lim_{x \rightarrow a} f(x) = L.$$

### Example 1.2

Consider the function  $f(x) = 2x + 3$ . We want to determine the limit of  $f$  at the point  $a = 1$ . We suspect that  $\lim_{x \rightarrow 1} f(x) = 5$ , since  $f(1) = 2 \cdot 1 + 3 = 5$ .

To show this, we take an arbitrary  $\varepsilon > 0$  and need to find a  $\delta > 0$ , such that for all  $x$  with  $0 < |x - 1| < \delta$  we have:  $|f(x) - 5| < \varepsilon$ .

Let's calculate  $|f(x) - 5|$ :

$$|f(x) - 5| = |(2x + 3) - 5| = |2x - 2| = 2|x - 1|.$$

To ensure that  $|f(x) - 5| < \varepsilon$ , we set  $2|x - 1| < \varepsilon$ , which is equivalent to  $|x - 1| < \frac{\varepsilon}{2}$ .

Therefore, we can choose  $\delta = \frac{\varepsilon}{2}$ . Then for all  $x$  with  $0 < |x - 1| < \delta$  we have:

$$|f(x) - 5| = 2|x - 1| < \varepsilon.$$

Thus, we have shown that  $\lim_{x \rightarrow 1} f(x) = 5$ .

### Exercise 1.3

Calculate the limit  $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$ .

**Proposition 1.4**

Let  $f : D \rightarrow \mathbb{R}$  be a function with  $D \subseteq \mathbb{R}$  and  $a$  a cluster point of  $D$ . If  $\lim_{x \rightarrow a} f(x) = L$  exists, then this limit is unique.

*Proof.* Assume that there exist two different limits  $L_1$  and  $L_2$  of  $f$  at the point  $a$ . This means that for all  $\varepsilon > 0$  there exists a  $\delta_1 > 0$ , such that for all  $x \in D$  with  $0 < |x - a| < \delta_1$  we have:  $|f(x) - L_1| < \varepsilon$ . Similarly, for all  $\varepsilon > 0$  there exists a  $\delta_2 > 0$ , such that for all  $x \in D$  with  $0 < |x - a| < \delta_2$  we have:  $|f(x) - L_2| < \varepsilon$ .

Now choose  $\varepsilon = \frac{|L_1 - L_2|}{3}$ . Then there exist  $\delta_1 > 0$  and  $\delta_2 > 0$ , such that for all  $x$  with  $0 < |x - a| < \delta_1$  we have:  $|f(x) - L_1| < \varepsilon$ , and for all  $x$  with  $0 < |x - a| < \delta_2$  we have:  $|f(x) - L_2| < \varepsilon$ .  $\square$

**Corollary 1.5**

If  $f : D \rightarrow \mathbb{R}$  is a function with  $D \subseteq \mathbb{R}$  and  $a$  a cluster point of  $D$ , and if  $\lim_{x \rightarrow a} f(x) = L$  exists, then we also have  $\lim_{x \rightarrow a} |f(x)| = |L|$ .

*Proof.* Since  $\lim_{x \rightarrow a} f(x) = L$  exists, for all  $\varepsilon > 0$  there exists a  $\delta > 0$ , such that for all  $x \in D$  with  $0 < |x - a| < \delta$  we have:  $|f(x) - L| < \varepsilon$ . We want to show that  $\lim_{x \rightarrow a} |f(x)| = |L|$ .

Let's calculate  $||f(x)| - |L||$ :

$$||f(x)| - |L|| \leq |f(x) - L| < \varepsilon.$$

Therefore, for all  $x$  with  $0 < |x - a| < \delta$  we have:  $||f(x)| - |L|| < \varepsilon$ . Thus, we have shown that  $\lim_{x \rightarrow a} |f(x)| = |L|$ .  $\square$

**Fact 1.6**

If  $f : D \rightarrow \mathbb{R}$  is a function with  $D \subseteq \mathbb{R}$  and  $a$  a cluster point of  $D$ , and if  $\lim_{x \rightarrow a} f(x) = L$  exists, then we also have  $\lim_{x \rightarrow a} (f(x) + c) = L + c$  for any constant  $c \in \mathbb{R}$ .

**Theorem 1.7**

Let  $f : D \rightarrow \mathbb{R}$  be a function with  $D \subseteq \mathbb{R}$  and  $a$  a cluster point of  $D$ . If  $\lim_{x \rightarrow a} f(x) = L$  exists, then we also have  $\lim_{x \rightarrow a} (c \cdot f(x)) = c \cdot L$  for any constant  $c \in \mathbb{R}$ .

*Proof.* Since  $\lim_{x \rightarrow a} f(x) = L$  exists, for all  $\varepsilon > 0$  there exists a  $\delta > 0$ , such that for all  $x \in D$  with  $0 < |x - a| < \delta$  we have:  $|f(x) - L| < \varepsilon$ . We want to show that  $\lim_{x \rightarrow a} (c \cdot f(x)) = c \cdot L$ .

Let's calculate  $|c \cdot f(x) - c \cdot L|$ :

$$|c \cdot f(x) - c \cdot L| = |c| \cdot |f(x) - L| < |c| \cdot \varepsilon.$$

$\square$

## Chapter 2

# Continuity

### Definition 2.1 Continuity

A function  $f : D \rightarrow \mathbb{R}$  with  $D \subseteq \mathbb{R}$  is continuous at the point  $a \in D$ , if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

A function is continuous if it is continuous at every point in its domain.





# Bibliography